

Hybrid Quantum–Classical Pauli-Correlator Architecture for Image Reconstruction

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Abstract

We introduce the Pauli-Correlator Autoencoder (PCAE), a hybrid quantum–classical architecture for image reconstruction. PCAE assembles matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy term, and a classical decoder into a single pipeline. Two instantiations, PCAE1D and PCAE2D, realise chain and grid interaction graphs. Group-averaged observable pooling makes the PCAE2D feature map exactly equivariant to the eight symmetries of the square. The contribution is this specific assembly rather than any ingredient in isolation. Across six real image datasets, per-image-trained PCAE2D models reconstruct with peak signal-to-noise ratio (PSNR) between 25.8 and 41.6 decibels. Five fitted checkpoints are replayed on superconducting quantum hardware without retraining, reaching 25.90 to 31.52 decibels. Under a fixed linear readout, an affine-readout argument and a synthetic test show that a pair-observable channel is necessary for targets built from distant-patch products. The learned Hamiltonian energy gives an interpretable readout of the trained representation. The results establish PCAE as a working hybrid architecture that fits varied images and runs on quantum hardware at the six-qubit scale.

Keywords: Quantum image processing, Tensor networks, Variational quantum algorithms, Pauli observables, Group-equivariant representations, Quantum hardware pilot

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1 Introduction

Hybrid quantum–classical models are now often proposed for vision tasks. The aim is to obtain representations with a favourable inductive bias. Such models can use entanglement, physically motivated regularisers, or tensor-network structure, which standard convolutional networks do not have (Cerezo et al. 2021; Tilly et al. 2022; Preskill 2018). The variational quantum eigensolver and its later variants use the fact that physical Hamiltonians split into local terms. This allows them to build shallow circuits suited to the problem (Cerezo et al. 2021; Tilly et al. 2022; Bauer et al. 2020). Quantum image processing gives several ways to store pixel values directly in qubits. These range from amplitude-style encodings (Le et al. 2011) to basis-encoded pixel values (Zhang et al. 2013). Basic image operations on these encodings have also been developed (Chakraborty et al. 2022; Zhou et al. 2017; Yao et al. 2017). Tensor-network methods work well as compressed representations of structured classical data, such as images (Cirac et al. 2021; Latorre 2005; Han et al. 2018; Stoudenmire and Schwab 2016; Ran et al. 2020; Guala et al. 2023) and particle-physics events (Araz and Spannowsky 2022). Hybrid models have also been applied to related imaging tasks, such as tomographic reconstruction (Haga 2024; Nau et al. 2023) and ghost imaging (Zhai et al. 2025). Quantum convolutional and capsule networks are further

examples (Slabbert and Petruccione 2024; Cong et al. 2019; Hur et al. 2022; Liu et al. 2023). PCAE focuses on the combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the Pauli-Correlator Autoencoder (PCAE1D/PCAE2D), which combines these ingredients into a single architecture with three properties studied jointly. The first is a *Pauli-correlator latent space*, in which measured VQC observables rather than a raw quantum state form the representation handed to a classical decoder, following quantum-enhanced feature maps (Havlíček et al. 2019). The second is an *exact, numerically verified D_4 -equivariant observable-pooling construction*, which group-averages the observable grid over the eight symmetries of the 4×4 patch grid rather than motivating symmetry through architecture choices alone. This follows group-equivariant classical vision (Cohen and Welling 2016; Bronstein et al. 2021) and the equivariant-quantum-neural-network line that formalises symmetry-embedding constructions for variational circuits (Meyer et al. 2023; Nguyen et al. 2024; Schatzki et al. 2024; West et al. 2024, 2025). The third is a *learnable Hamiltonian energy term*, used as an auxiliary training objective and read out as an interpretable diagnostic, with full nearest-neighbour $XX/YY/ZZ$ sectors in PCAE1D and grid-edge ZZ terms in PCAE2D. Table 10 contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

Novelty relative to the nearest hybrid-vision architectures

PCAE’s closest architectural relatives are three hybrid quantum-vision patterns already established individually in the literature.

Register-based quantum autoencoders (Romero et al. 2017; Wang et al. 2024; Asaoka and Kudo 2025) encode an image’s pixel amplitudes directly into a qubit register and train a trash-qubit fidelity objective. PCAE feeds the circuit only quantum-inspired, pre-compressed MPS patch features, not raw pixel amplitudes. It hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score. PCAE is an autoencoder in the usual encoder–decoder sense. It maps each image patch to a measured observable vector and decodes it back to pixels. It does not compress a quantum state.

Quantum-kernel vision models (Havlíček et al. 2019; Beaulieu et al. 2022) embed an input into a quantum feature space and read out a scalar kernel value consumed by a classical kernel method. PCAE reads out an explicit vector of local and pairwise Pauli expectation values as a latent representation for a trained decoder, and adds enforced group-equivariant pooling and a Hamiltonian energy term that a kernel construction has no analogue for.

Tensor-network (TN) circuit classifiers (Guala et al. 2023; Stoudenmire and Schwab 2016; Han et al. 2018) use MPS structure either as a standalone classical model or as a template for a shallow quantum circuit trained for classification. PCAE uses MPS compression only as a quantum-inspired *preprocessing* step that sets a separately trained VQC’s rotation angles, and targets per-image reconstruction rather than classification.

PCAE’s contribution is the assembly of MPS preprocessing and the three properties above in one pipeline (Table 10). Section 5.2 examines how the individual ingredients recur across this literature.

Contributions

1. **Architecture.** PCAE1D/PCAE2D combines MPS patch features, a six-qubit VQC, a 19–27-dimensional Pauli-correlator vector, 2D Fourier positional encoding, a graph-Hamiltonian energy term, and a classical decoder.
2. **Reconstruction scope (per-image fitting).** Identically configured per-image fitting experiments cover six real image datasets (Section 4.1).
3. **Held-out surrogate comparison.** On held-out CIFAR-10, a closed-form surrogate of the PCAE2D observable map meets the ± 1 dB equivalence criterion against four of seven resource-matched classical/ablated feature maps, under two one-sided tests (TOST) with Holm correction. It clearly outperforms the two random-feature null controls. The trained circuit itself stays near the random-latent range on held-out images at the tested budget (Section 4.3).
4. **Hardware pilot.** Five fitted checkpoints are replayed on IBM hardware without retraining and decoded from hardware-measured observables (Section 4.4), strengthened by a noise/shot-budget sweep and repeated execution across backends (Sections 4.5–4.6).
5. **Equivariant observable pooling (proposition).** An exact, numerically verified D_4 -equivariant observable-pooling construction makes the PCAE2D observable map equivariant to the eight symmetries of the square, to numerical precision (Section 4.7). The guarantee follows from Reynolds group averaging over the D_4 orbit rather than from anything quantum, and the same construction gives the same exactness for a purely classical feature map. We report it as a design property of the pipeline.
6. **Restricted nonlocal result (diagnostic).** In a leakage-audited protocol with a fixed linear readout, pair observables are necessary when the target depends explicitly on distant-patch products (Section 4.8).

The Supplementary Material gives the complete held-out surrogate study, the hardware job identifiers and validation data, and provenance audits of excluded exploratory results.

2 Pauli-Correlator Autoencoder: Architecture

PCAE treats each spatial patch as a finite spin state and compresses it as a matrix product state (MPS). A variational quantum circuit (VQC) then measures a fixed set of Pauli observables, which a classical multi-layer perceptron (MLP) decodes back into pixels. A learnable graph-Hamiltonian energy term regularises training. Parameterised Hamiltonian learning approaches treat the Hamiltonian as the primary learned object (Shi et al. 2023); in PCAE it is instead an auxiliary objective alongside the reconstruction loss, and its learned energy is read out as a diagnostic.

Problem statement.

Given an image x partitioned into non-overlapping patches $\{x_p\}_{p=1}^P$, PCAE learns a per-patch reconstruction map. Each patch is compressed to a quantum-inspired MPS feature vector $\phi_{\text{MPS}}(x_p)$, a learned linear layer sets the VQC rotation angles, and measuring a fixed set of Pauli observables yields the correlator feature map $\Phi(x_p) \in \mathbb{R}^d$ ($d = 27$ for PCAE1D, 19 for PCAE2D). With $n = 6$ qubits, bond set $\mathcal{E}$, and c measured pair channels per bond,

$$d = 2n + c|\mathcal{E}| = \begin{cases} 2(6) + 3(5) = 27, & \text{PCAE1D (ZZ, XX, YY on a 5-bond chain),} \\ 2(6) + 1(7) = 19, & \text{PCAE2D (ZZ on 7 grid edges),} \\ 2(6) + 1(5) = 17, & \text{ZZ-only ablation control.} \end{cases} \quad (1)$$

All PCAE1D and PCAE2D circuit results use $d = 27$ and $d = 19$. A classical decoder D_θ maps $\Phi(x_p)$, together with a Fourier positional encoding $\pi(p)$, back to a reconstructed patch $\hat{x}_p = D_\theta(\Phi(x_p), \pi(p))$. Training minimises the pixel-reconstruction loss with the graph-Hamiltonian energy term $E(\cdot)$ as an auxiliary regulariser,

$$\mathcal{L}(\theta) = \frac{1}{Pm} \sum_{p=1}^P \|x_p - \hat{x}_p\|_2^2 + \lambda \frac{1}{P} \sum_{p=1}^P E(\Phi(x_p)), \quad (2)$$

where m is the number of pixels per patch, so the first term is the mean squared pixel error, and the second term is the mean graph-Hamiltonian energy over patches and λ sets its weight (Section 5.1 reports its effect). The energy of patch p is

$$E(\Phi(x_p)) = \sum_{(u,v) \in \mathcal{E}} \sum_{\alpha \in S} J_{uv}^\alpha \langle \sigma_u^\alpha \sigma_v^\alpha \rangle_p, \quad (3)$$

where $\mathcal{E}$ is the bond set of Eq. (1), $S = \{x, y, z\}$ for PCAE1D and $S = \{z\}$ for PCAE2D, and the couplings J_{uv}^α are learned. The shared pipeline is

$$\begin{aligned} \text{Image} &\rightarrow \{\text{Patches}\} \rightarrow \text{MPS} \\ &\rightarrow 46\text{-D}/22\text{-D Feature} \rightarrow 6\text{-D Vector} \\ &\rightarrow 27\text{-D}/19\text{-D Latent} \rightarrow \text{Decoder} \\ &\rightarrow \text{Reconstruction.} \end{aligned} \quad (4)$$

Why MPS features.

Treated as an amplitude-encoded quantum state, a patch of n sites has a coefficient tensor with 2^n entries. Matrix-product-state factorisation instead represents it as a chain of small tensors bounded by a fixed bond dimension χ . This is a compact, singular-value-controlled compression whose entanglement-entropy profile is already known to track image compressibility (Latorre 2005; Stoudenmire and Schwab 2016), and whose local expectations and correlations are cheap features that do not require a full state-preparation circuit.

Table 1 The two executed configurations. Both use six qubits and the same decoder; they differ in patch size and in the measured observable set.

	PCAE1D	PCAE2D
Benchmark	<i>Monalisa</i>	multi-dataset
Image size	256×256	32×32 (native)
Patch size	64×64	8×8
MPS sites	12	6
Local X/Z expectations	24	12
Nearest-neighbour ZZ	11	5
Bipartite entropies	11	5
MPS feature vector	46-D	22-D
Measured observables	27-D	19-D
local Z , local X	6, 6	6, 6
pair terms	5 each ZZ, XX, YY	7 grid-edge ZZ
Fourier positional vector	8-D	4-D

Why a Pauli-correlator latent space rather than a raw quantum state.

Handing the classical decoder a fixed vector of measured Pauli expectation values, instead of full state tomography, keeps the quantum-to-classical interface at a size linear in the qubit count, on the bounded-degree graphs used here, rather than exponential readout cost. The execution cost also grows with the number of measurement settings and the shots per setting. It also follows the measured-observable-as-feature pattern used by quantum-enhanced feature maps generally (Havlíček et al. 2019). This choice is what makes the interpretability diagnostics of Section 5.1 and the training-dynamics diagnostics of Section 4.9 possible, since a full state vector would not expose a comparably compact, physically meaningful summary.

Why a grid rather than a chain interaction topology for PCAE2D.

Aligning the six-qubit interaction graph with the 2×3 patch-neighbour adjacency, rather than an arbitrary linear chain, gives the latent a two-dimensional coupling pattern. The D_4 pooling of Section 4.7 does not depend on this choice. It acts by permuting the 4×4 grid of patches in an image, and a 2×3 graph has no 90° rotation symmetry of its own. PCAE1D’s chain topology is the control without a grid. It is used to measure what the grid adds (Section 5.3).

Two preprocessing configurations were executed, one per topology. Both follow the same three stages. Each image is first divided into sixteen non-overlapping patches. Every patch is then ℓ_2 -normalised and sequentially SVD-compressed to a matrix product state of bond dimension $\chi = 4$. Finally, the local expectations, nearest-neighbour correlations and bipartite entropies of that state are read off as the quantum-inspired MPS feature vector. The two configurations differ only in the sizes involved, which Table 1 lists in full. The 2^n pixel values of a patch ($n = 6$ for 8×8 , $n = 12$ for 64×64) are flattened in row-major order and reshaped into an order- n tensor with binary indices, so each index is one site of dimension 2. The features are expectation values

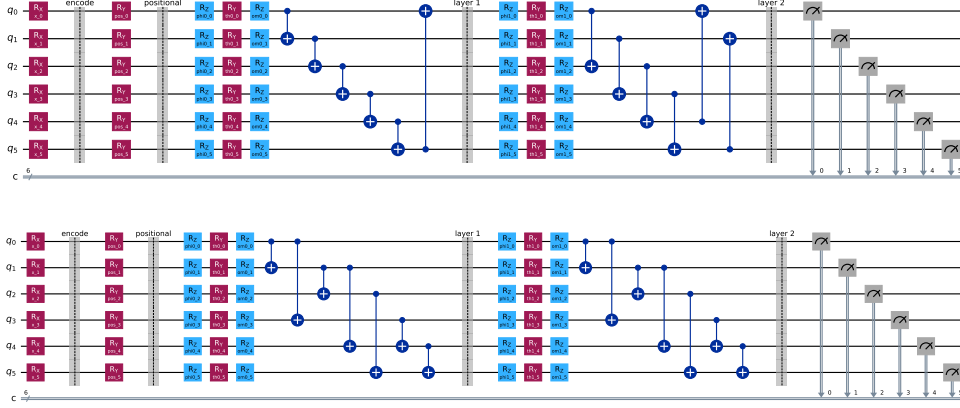


Fig. 1 The PCAE1D (top) and PCAE2D (bottom) variational ansätze, exactly as executed (encoding R_X rotations, positional R_Y rotations, then two entangling layers of R_Z - R_Y - R_Z single-qubit rotations with a CNOT ring for PCAE1D or fixed grid-edge CNOTs for PCAE2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section 4.4.

and Schmidt spectra of the compressed state, so they do not depend on its canonical form. In the multi-dataset runs of Section 4.1, 10^{-4} is added to every pixel before normalisation, so that all-zero patches, such as MNIST background, can be normalised. The remaining stages are common to both. A learned linear layer compresses the MPS feature vector to six circuit angles, one per qubit. After the circuit is executed, a two-hidden-layer MLP (128 and 256 units) maps the measured observables together with the Fourier positional vector back to a single greyscale patch. The positional encoding follows established use as a spatial inductive bias in classical generative vision models (Xu et al. 2021). Fourier features of this kind also help coordinate-based networks fit fine image detail (Tancik et al. 2020). During training the circuit is simulated exactly, so the decoder receives exact expectation values. Finite-shot estimates enter only in the hardware replay and the noisy simulation of Sections 4.4–4.6. Figure 1 shows the exact PCAE1D and PCAE2D ansatz circuits. These were rebuilt in Qiskit for the hardware pilot of Section 4.4, and numerically verified against the training-time PennyLane circuit before use.

Table 2 summarises the tested variants. PCAE1D uses a linear 6-qubit chain (5 nearest-neighbour bonds, 27-D observable vector); PCAE2D arranges the same 6 qubits on a 2×3 lattice matching patch-boundary adjacency (19-D latent signature); a symmetric PCAE1D variant adds a $U(1)$ -preserving coupling constraint; and a D_4 -pooled PCAE2D variant (Section 4.7) group-averages the observable grid over the eight symmetries of the 4×4 patch grid, $\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx)$, which is equivariant by construction. Both PCAE1D and PCAE2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

Table 2 PCAE variants tested in this paper.

Variant	Topology	Latent dim.	Symmetry
PCAE1D	6-qubit chain	27-D	None (control)
PCAE2D	2×3 grid	19-D	D_4 -motivated only
Symmetric PCAE1D	6-qubit chain	27-D	U(1) coupling
D_4 -pooled PCAE2D	2×3 grid, pooled	19-D	D_4 -equivariant (exact)

3 Experimental Setup

The reconstruction results in Sections 4.1, 4.2 and 4.4–4.6 use the single-image *Monalisa* protocol developed for PCAE. A fresh model is optimised directly on its target image, with no held-out split. These results therefore measure per-image fitting and reconstruction capability. The hardware pilot replays the resulting fitted checkpoints on real devices without retraining. The symmetry and restricted pair-observable diagnostics use closed-form feature maps and the separate protocols stated in their respective sections. The training-dynamics readouts of Section 4.9 are reported as an interpretability diagnostic, with their scope set by the negative control given there. Held-out generalisation and component necessity are evaluated under the resource-matched statistical protocol, using a closed-form surrogate of the PCAE2D observable map (Section 4.3). All settings report MSE and $\text{PSNR} = 20 \log_{10}(1/\sqrt{\text{MSE}})$; image settings additionally report SSIM. Full implementation settings and environment versions are given in the Supplementary Material.

4 Results

4.1 Reconstruction across six real image datasets

For each image, we trained a fresh PCAE2D model using non-overlapping 8×8 patches (16 patches/image, 80–100 optimisation steps). The sample covers CIFAR-10 (Krizhevsky 2009), MNIST (LeCun et al. 1998), Fashion-MNIST (Xiao et al. 2017), and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 (Yang et al. 2023). Table 3 reports mean PSNR between 25.8 and 41.6 dB (SSIM 0.91–1.00). This establishes that the architecture fits varied image structures under a common protocol. This protocol is per-image fitting, in the same sense as implicit neural representations, which fit one network to one image or scene (Ulyanov et al. 2020; Sitzmann et al. 2020; Tancik et al. 2020; Mildenhall et al. 2020; Dupont et al. 2021; Strümpfer et al. 2022). Quantum versions of this idea fit a single signal or image with a variational circuit (Zhao et al. 2024; Fujihashi and Koike-Akino 2024; Lizzio Bosco et al. 2026). The closest of these, QIREN, is a quantum coordinate network aimed at a fitting advantage over classical Fourier networks (Zhao et al. 2024). PCAE is instead a patch autoencoder with a measured-correlator bottleneck, and it aims at a practical hybrid pipeline whose quantum part is the measured observable vector. These results therefore measure representation capacity. Generalisation to held-out images is tested

Table 3 PCAE2D same-set reconstruction across six real image datasets (per-image trained, 80–100 steps, mean $\pm$ std over sampled images; SSIM is computed globally over each image).

Dataset	n images	PSNR (dB)	SSIM
CIFAR-10	3	37.75 ± 2.38	0.9986 ± 0.0003
MNIST	3	25.77 ± 2.23	0.9783 ± 0.0116
Fashion-MNIST	3	34.55 ± 3.05	0.9980 ± 0.0006
PathMNIST	3	36.86 ± 2.11	0.9118 ± 0.0627
BloodMNIST	2	36.57 ± 2.36	0.9922 ± 0.0063
PneumoniaMNIST	2	41.60 ± 0.98	0.9985 ± 0.0006

in Section 4.3. A compression claim would also need a bit rate, which we do not report (Dupont et al. 2021; Strümpler et al. 2022).

4.2 Scope of per-image fitting

A model optimised on one image is specific to that image. On *Monalisa*, zero-shot transfer to a second image reaches 8.31 dB PSNR, with SSIM 0.1044. Extending training to include the second image recovers 28.63 dB, with SSIM 0.9858. The per-image results above therefore measure per-image optimisation and reconstruction capability. Section 4.3 tests the held-out case directly.

4.3 Held-out comparison with a closed-form surrogate

The results above measure per-image fitting. A separate question is whether PCAE’s feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol. This comparison uses a closed-form surrogate of the PCAE2D observable map, so that many seeds and controls can be run. The surrogate combines 18 local patch statistics, six products of local-feature pairs that stand in for the pairwise ZZ correlators, the sines of those products, and positional features. It does not evaluate the circuit. The main results are summarised below, including the equivalence tests of Table 4.

On held-out CIFAR-10, the raw/local linear control reaches 18.80 ± 1.42 dB, against the PCAE2D surrogate’s 18.12 ± 1.54 dB. Aggregating held-out images within each split gives a PCAE2D-minus-control difference of -0.68 dB, with bootstrap 95% CI $[-0.91, -0.46]$ dB and Wilcoxon $p = 0.0625$ at $n = 5$ seeds. The same numerical direction holds on five further real datasets. The surrogate map is therefore informative. Its mean stays within 1 dB of the simpler maps, and the raw/local linear control is 0.68 dB ahead of it.

All controls are matched at the readout. Each passes a 32-D feature vector to the same linear readout with 2112 parameters, on the same splits. This matches the readout resources. It does not match the feature class. The controls differ in what their features can express. The no-entanglement and ZZ -only controls remove all or half of the pair products, and the quadratic classical control adds second-order classical terms. The surrogate is a 32-D map, not the 19-D circuit map of Table 1. The quadratic

Table 4 TOST equivalence test on held-out CIFAR-10 (± 1 dB margin, $\alpha = 0.05$, five seed-level paired differences, closed-form PCAE2D surrogate minus control). The verdict uses the Holm-adjusted p . Without correction, equivalence is the same as the 90% CI lying inside $[-1, +1]$ dB. “No entanglement” removes the pair products, and “Random VQC” is a block of Gaussian random features.

Control	Δ PSNR (dB)	90% CI (dB)	TOST p	Holm p	Verdict
Raw/local linear	-0.68	$[-0.96, -0.40]$	0.034	0.103	Not equivalent
No entanglement	-0.34	$[-0.69, +0.02]$	0.008	0.032	Equivalent
ZZ-only observables	-0.44	$[-0.61, -0.27]$	0.001	0.007	Equivalent
Quadratic classical	-0.21	$[-0.46, +0.03]$	0.001	0.007	Equivalent
Shuffled pair observables	-0.19	$[-0.35, -0.02]$	< 0.001	0.002	Equivalent
Strict classical random features	+2.27	$[+1.92, +2.63]$	0.999	1.000	PCAE2D better
Random VQC	+6.97	$[+5.56, +8.39]$	1.000	1.000	PCAE2D better

classical control augments the 18 local features with the nine products of adjacent local-feature pairs and their sines, together with the positional features, selected to the same 32-D width. It therefore has the same form as the surrogate, with nine products in place of six.

A TOST equivalence test (Schuirmann 1987; Lakens 2017) at a declared ± 1 dB margin asks whether each gap is small enough to call two maps equivalent rather than merely “not shown to differ”. The estimand is the seed-level PCAE2D-minus-control PSNR difference, averaged over held-out images within each seed. Without correction, equivalence requires both one-sided tests to reject at $\alpha = 0.05$. This is the same as the 90% confidence interval lying inside $[-1, +1]$ dB (Table 4). The margin is a declared choice. With five seeds, $p = 0.0625$ is the smallest attainable nonzero two-sided Wilcoxon signed-rank p -value, so that test cannot reject at the 0.05 level. The t -based 90% interval is wider than the percentile-bootstrap 95% interval in these data; bootstrap intervals can be unstable with only five seed-level observations. For scale, a 1 dB PSNR difference corresponds to a factor of approximately $10^{0.1} \simeq 1.26$ in MSE. After Holm correction across the seven comparisons, four controls meet the equivalence criterion against PCAE2D within that margin: no entanglement, ZZ-only observables, quadratic classical features, and shuffled pair observables. The raw/local linear control meets the uncorrected criterion ($p = 0.034$) but not the Holm-corrected criterion (adjusted $p = 0.103$). The two weakest controls, strict classical random features and random-VQC, are not equivalent because the surrogate substantially outperforms them. The trained circuit has been tested on held-out images in the topology comparison of Section 5.3. Trained on two images for 90 steps, PCAE2D reaches 11.57 ± 1.53 dB on three unseen CIFAR-10 images, close to the 11–15 dB random-latent range (Section 4.4). Held-out generalisation of the trained circuit is therefore not established at this budget.

We report this as a complement to the positive pair-observable and symmetry results below. It delineates their tested boundary and records the provenance checks that prevented exploratory artifacts from becoming claims.

A matched-budget, multi-seed rerun of the component-ablation table reaches the same conclusion on ordinary per-image reconstruction. The rerun uses 240 steps and 3 seeds per variant. A plain classical-replacement control ties or slightly exceeds the quantum baseline, and only a circuit-free random-VQC control performs clearly worse. This rerun uses a descriptive pooled-standard-deviation comparison rule at 3 seeds rather than a calibrated test, and full results are in the Supplementary Material. Section 5.2 discusses why this ties-with-classical result and the pair-observable necessity result of Section 4.8 are consistent.

4.4 A real hardware reconstruction pilot

The results above are noiseless-simulator results. We now report a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor).

The pilot covers five images. These are the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship* $\times 2$, *airplane*), each optimised directly on its own target image following the same per-image protocol as Sections 4.1 and 4.2. For each image we took an already-trained PCAE1D/PCAE2D checkpoint’s exact learned parameters and rebuilt its measurement circuits gate-for-gate in Qiskit. We verified the translation against the original PennyLane simulator to within numerical noise, at most 7.9×10^{-4} maximum absolute error per observable across every patch. We then executed the same circuits on real hardware with 256 shots per measurement basis. This is a fixed per-basis shot budget, rather than an adaptive shadow-tomography-style estimation scheme (Huang et al. 2020). The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

Reconstructions from real, noisy hardware measurements retain recognisable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2). The four CIFAR-10 images average 29.56 ± 2.03 dB on hardware against 42.69 ± 3.84 dB in simulation (Figure 3, Table 5). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. Circuits were transpiled at optimisation level 1 and submitted through SamplerV2 with default options; dynamical decoupling, Pauli twirling and readout-error mitigation were not enabled. Each image used 256 shots per basis, totalling 12,288 shots for *Monalisa* and 8,192 for each CIFAR-10 image. The original jobs’ physical-qubit layouts and contemporaneous calibration timestamps were not serialised and could not subsequently be recovered. Every hardware reconstruction remains above the random-latent control range (11–15 dB). The pilot used 176 circuits; the contemporaneous account-meter reading increased by 16 quantum-processor seconds.

The pilot establishes that the trained PCAE observable channel survives execution on present-day hardware. With five images it is a pilot, and it makes no hardware-advantage claim. The learned parameters, the measurement circuits and the classical decoder all transfer unchanged, and the decoded images remain quantitatively informative at 256 shots per basis on an unmitigated device. This is the sense in which

Table 5 Real IBM hardware reconstruction pilot (backend `ibm_fez`, 256 shots/basis, no retraining). PSNR compares hardware-decoded and simulator-decoded reconstructions against the same ground-truth image.

Image	Patches	Sim. PSNR (dB)	HW PSNR (dB)
Monalisa (PCAE1D)	16	37.99	25.90
CIFAR cat (PCAE2D)	16	44.85	31.52
CIFAR ship, hydrofoil (PCAE2D)	16	36.56	26.44
CIFAR ship, sea boat (PCAE2D)	16	42.57	31.20
CIFAR airplane (PCAE2D)	16	46.79	29.10
CIFAR mean $\pm$ std (4 images)	—	42.69 ± 3.84	29.56 ± 2.03



Fig. 2 *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint’s noiseless-simulator replay (centre) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognisable.

PCAE runs on real quantum hardware. Gate sequences, measurement construction, job identifiers and validation results are in the Supplementary Material.

The same checkpoint-replay methodology supports a further hardware replay. Fresh PCAE1D checkpoints were trained at $N \in \{4, 6, 8\}$ and replayed epoch-by-epoch on real IBM Quantum hardware and on the IonQ cloud simulator (IonQ 2026). That exercise serves the training-dynamics diagnostic of Section 4.9. It is a single-patch, single-seed probe, reported in the Supplementary Material as an interpretability readout.

4.5 Noise/shot trade-off: a free simulator complement

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. To use IBM Quantum time well, we separate what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (FakeFez, no quantum-processor time consumed), we replay the same already-trained checkpoints’ circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions

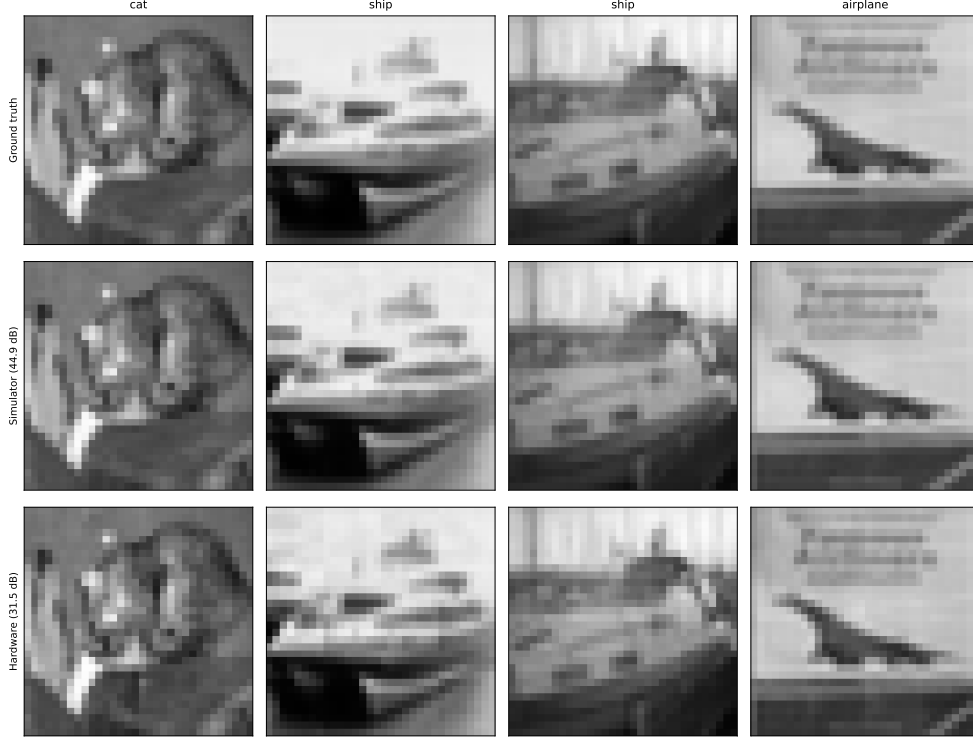


Fig. 3 Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained PCAE2D checkpoints (middle row) and ground truth (top row).

each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap. Moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots do not reliably recover it. Increasing to 4096 shots per basis changes PSNR by only -0.05 dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a 0.96 dB loss. At this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality.

Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture. That comparison leaves a further 4.24 dB average gap, ranging from 2.25 to 8.06 dB across images. This residual is consistent with effects that the calibration model does not capture, such as drift between calibration and execution, crosstalk, and correlated errors. The pilot does not separate these sources. The airplane image is the clear outlier, with an 8.06 dB residual gap, roughly double any other image. We report it rather than exclude it, since a single-execution, five-image pilot cannot distinguish an image-specific effect from

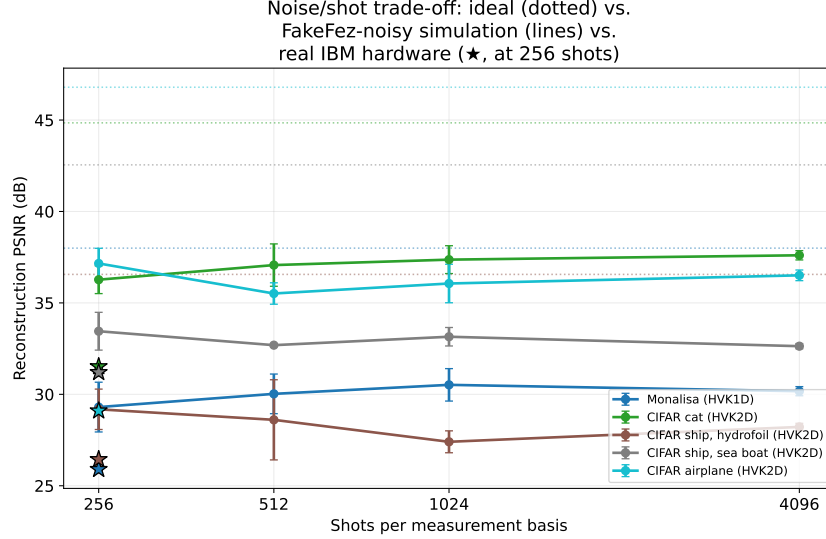


Fig. 4 Calibrated device-noise simulation (FakeFez, mean $\pm$ std over three executions). Dotted lines show noiseless ceilings and stars show real-hardware measurements. Increasing 256 to 4096 shots changes mean PSNR by only -0.05 dB.

ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it. It bounds how much of the observed hardware gap is attributable to modelled device noise rather than unmodelled effects, using zero additional quantum-processor time.

4.6 Real-hardware anchor points: repeated execution and a second backend

The simulated trade-off above is useful only if it reflects real devices. We checked this directly on hardware with four jobs from the free-tier IBM Quantum allowance. These four jobs used well under 1% of the monthly limit. Two checkpoints, *Monalisa* and *CIFAR cat*, were replayed with no retraining on *ibm_marrakesh* and *ibm_kingston*. Both backends are distinct from the original pilot’s *ibm_fez*. We ran them at the original 256-shot budget and, separately, at 1024 shots.

Monalisa reproduces almost exactly across backends at matched shots (25.94 vs. 25.90 dB, *ibm_marrakesh* vs. *ibm_fez*), and gains only 0.16 dB from quadrupling shots to 1024, consistent with the simulated finding above that shot count is not a reliable lever. The *CIFAR cat* point is the more informative one. On *ibm_kingston* at the original 256-shot budget it reaches 28.93 dB, which is 2.59 dB below the *ibm_fez* pilot point. Its own 1024-shot execution on the same backend recovers to 31.24 dB, within 0.28 dB of the original. This is real job-to-job and backend-to-backend variation of a magnitude comparable to the unmodelled residual identified by the simulator comparison (Section 4.5). It gives direct evidence that repeated execution matters at

Table 6 Real-hardware anchor points: same checkpoints, no retraining, on two further backends. The original pilot’s `ibm_fez` point is shown for reference.

Image	Backend	Shots	PSNR (dB)
Monalisa	<code>ibm_fez</code> (original)	256	25.90
Monalisa	<code>ibm_marrakesh</code>	256	25.94
Monalisa	<code>ibm_kingston</code>	1024	26.10
CIFAR cat	<code>ibm_fez</code> (original)	256	31.52
CIFAR cat	<code>ibm_kingston</code>	256	28.93
CIFAR cat	<code>ibm_kingston</code>	1024	31.24

least as much as backend choice or shot count. A single 256-shot run on one backend is not yet a stable estimate of hardware performance.

4.7 Exact D_4 -equivariant observable pooling

A D_4 -pooled PCAE2D observable-grid map is equivariant to the eight symmetries of the square *by construction*:

$$\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx), \quad \Phi_{D_4}(hx) = P_h \Phi_{D_4}(x) \quad \forall h \in D_4, \quad (5)$$

Here D_4 acts on an image x by rotating or reflecting its 4×4 grid of patches, and P_g permutes the per-patch observable vectors in the same way, so that $P_{gh} = P_g P_h$. The property then holds exactly for any base map Φ . Substituting hx and re-indexing the sum with $g' = gh$ gives $P_g^{-1} = P_h P_{g'}^{-1}$, and hence $\Phi_{D_4}(hx) = P_h \Phi_{D_4}(x)$. The numerical check below therefore tests the implementation, not the property. We run it on the closed-form surrogate feature map over 7000 patch transforms of cached CIFAR images (Table 7, Figure 5). The pooled map’s equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$, which is floating-point noise. The unpooled positional map has an error of 0.815 ± 0.129 , and the local/raw control has 0.837 ± 0.131 . Both are of order one. The patch grid’s geometry therefore makes the unpooled map *D_4 -motivated* only, and the explicit pooling construction is what delivers the property. This construction operates at the feature-grid level and serves here as a numerical diagnostic, rather than as a component of the trainable reconstruction pipeline. Integrating it end-to-end (Section 5.4) is concrete future work. Applied to the trained circuit, the same pooling reduces the equivariance error to order 10^{-8} . This route to symmetry differs from equivariant quantum neural networks, in which every circuit layer is built to commute with the group action (Nguyen et al. 2024). A recent image classifier of that kind builds C_4 rotation symmetry into the circuit itself (San Sebastian Sein et al. 2025). In PCAE the symmetry comes from averaging the observable grid over the group after measurement. The circuit itself is not equivariant, and the same average applies equally to a classical feature map.

Table 7 D_4 equivariance error for unpooled and pooled observable-grid maps (lower is better; the pooled map is group-averaged over all eight square symmetries; closed-form surrogate map).

Model	Mean error	Std. dev.
D_4 -pooled PCAE2D	9.57×10^{-17}	1.26×10^{-17}
No positional encoding	7.40×10^{-1}	1.52×10^{-1}
PCAE2D positional	8.15×10^{-1}	1.29×10^{-1}
Local/raw	8.37×10^{-1}	1.31×10^{-1}

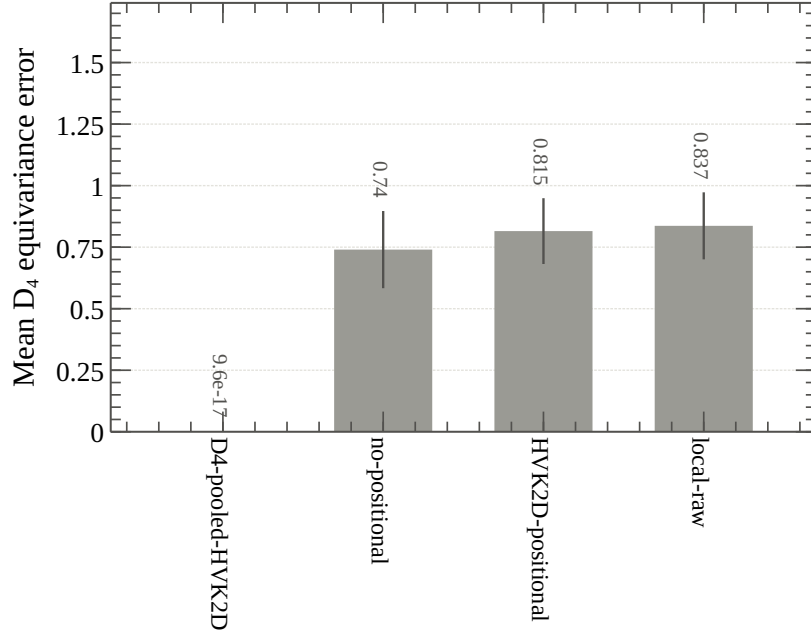


Fig. 5 D_4 equivariance diagnostic. The group-pooled map is equivariant to numerical precision. The unpooled positional map is not.

4.8 Pair-observable necessity in the nonlocal-structure protocol

The diagnostic tests whether a pair-observable channel is needed to fit a target that depends on pair correlations. The target is a fixed synthetic function of *simulated* patch features, and those features are not concatenated into any candidate’s own input block. Every control receives the same raw inputs as PCAE2D. All feature maps in this diagnostic are closed-form, and no circuit is evaluated. The pair-observable channel is represented by the analytic pair products that a pair measurement would supply. By construction, these products match the pair terms of the target. The experiment

Table 8 PCAE2D restricted pair-correlation diagnostic across five seeds. Deliberately favourable to pair-observable structure; evidence for pair-observable necessity under a matched linear readout. All features are closed-form. “Random VQC” is a block of Gaussian random features.

Model	Mean MSE	Mean PSNR (dB)	Mean R^2
PCAE2D pair observables	8.5969×10^{-4}	32.77	0.9735
Freeze quantum	4.7904×10^{-3}	27.34	0.8533
No pair observables	3.1461×10^{-2}	15.03	0.0191
Raw-linear classical	3.1654×10^{-2}	15.00	0.0133
Parameter-matched classical	3.3033×10^{-2}	14.82	-0.0297
Random VQC	3.7616×10^{-2}	14.25	-0.1732
Freeze classical	6.0973×10^{-2}	12.17	-0.9025

demonstrates representational capability under a deliberately aligned protocol, not a general performance advantage.

Table 8 and Figure 6 report the result across five seeds. PCAE2D’s pair-observable channel reaches mean $R^2 = 0.9735$, or 32.77 dB PSNR. This is a 17.74 dB improvement over a no-pair-observables control ($R^2 = 0.019$) and 18.52 dB over Gaussian random latents (“random VQC”) ($R^2 = -0.173$). Randomly rescaling and masking the pair features (“freeze quantum”), with only the readout trained, still reaches $R^2 = 0.853$. A raw-linear classical control ($R^2 = 0.0133$), a parameter-matched classical control ($R^2 = -0.030$), a random-feature control ($R^2 = -0.173$), and a frozen-classical control ($R^2 = -0.903$) all fail on this deliberately pair-dependent target. The no-pair-observables control retains only single-site nonlinear channels and removes every cross-site product feature. Under the fixed linear readout, it therefore cannot construct the cross terms on which the target depends.

The mechanism is explicit under the fixed linear readout. A linear map of single-patch expectations $\{\langle O_i \rangle\}$ cannot form the cross-term $\langle O_i \rangle \langle O_j \rangle$ that the target depends on. The pair measurement supplies the joint correlator $\langle O_i O_j \rangle$ directly as a feature. Formally, let $z_i = \langle O_i \rangle$ and let every admissible predictor be affine, $f(z) = w^\top z + b$. Then $\partial^2 f / \partial z_i \partial z_j = 0$ for all i, j . A target with a term $c z_i z_j$, $c \neq 0$, has this mixed derivative equal to c wherever z_i and z_j vary independently, so no affine f can fit it. Adding $z_i z_j$ to the feature vector makes the term linearly representable. A pair-observable channel is therefore *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target. A classical model that forms the same products, for example through quadratic features or a nonlinear readout, would also fit this target. For this reason we scope the claim to the linear protocol rather than to classical models in general.

4.9 A scoped training-dynamics diagnostic

Beyond the results above, we tracked two exploratory training-dynamics readouts. These are a magnetisation-style statistic $M_z(t)$ and a signed energy-to-entanglement ratio $R_{ES}(t)$. Both drift smoothly over training with run-to-run fluctuations, and

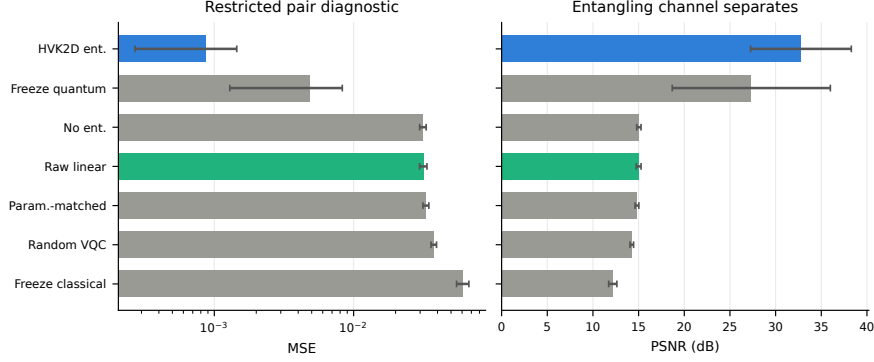


Fig. 6 Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The pair-observable feature map dominates every control on this deliberately favourable task.

neither shows a sharp transition. Any apparent onset is a within-run median-plus- 2σ threshold crossing, rather than a calibrated statistical test.

A negative control fixes the scope. We applied the identical protocol to a classical-replacement variant, in which the Hamiltonian energy is identically zero throughout training. That variant crosses the threshold in all 4/4 tested cases, which is as often as the Hamiltonian-regularised model’s 4/4. The threshold therefore fires whether the Hamiltonian term is present or absent, and it does not track that term. We therefore report these traces as an interpretability readout on the learned energy and entanglement.

5 Discussion

5.1 Effect of the Hamiltonian energy term

In this single-seed comparison, the signed Hamiltonian-energy objective substantially reduces reconstruction quality: removing it increases PSNR from 38.63 to 44.56 dB (Table 9). The legacy objective adds a signed mean energy to the reconstruction loss. Because learned energy can go negative, the optimiser can lower total loss by driving energy down without improving reconstruction. A contrastive variant, assigning lower energy to correct feature-position pairs than to shuffled pairs, recovers part of the gap (42.92 dB) but stays below its no-energy counterpart. These are fresh same-environment measurements (*Monalisa*, 200 steps, seed 42), and they supersede earlier values produced by a script that cannot be audited.

The learned couplings do not enter the decoder directly, but the energy term contributes gradients during training and therefore changes the learned circuit representation. It is consequently an auxiliary training objective as well as a post-training diagnostic. A physically motivated regulariser has room to help when it resolves an otherwise underdetermined objective. Under this protocol, the reconstruction loss alone provides the stronger optimisation signal.

Table 9 Hamiltonian-sector controls (single-seed, fresh same-environment reproduction: *Monalisa*, 200 steps, seed 42). The energy objective is diagnostically useful but does not improve reconstruction accuracy.

Experiment	PSNR (dB)	Interpretation
Legacy baseline, signed energy	38.63	Standard energy-regularised run
No energy loss	44.56	Removing energy improves PSNR
Contrastive Hamiltonian core	42.92	Stronger objective helps modestly vs. baseline, still below no-energy-loss

The training-dynamics readouts of Section 4.9 provide a descriptive window into how the learned energy and entanglement evolve. They are descriptive readouts rather than calibrated statistical tests, and the thermodynamic reading is outside their scope.

5.2 Task-dependent representational scope

Whether a quantum feature map offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class. Dequantisation results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched once the right classical feature is identified (Huang et al. 2021; Tang 2019). Where provable separations do exist, they are established for carefully constructed learning problems rather than for generic structured data (Gyurik and Dunjko 2023). We separate three questions. First, PCAE at six qubits can be simulated exactly on a classical computer, which says little about larger scales. Second, whether its model class admits an efficient classical replacement at scale is open. Tensor-network analyses give conditions for such dequantisation of variational models (Shin et al. 2024), and a classical tie at small scale does not settle it. Random Fourier features can also replace some variational models efficiently, under stated conditions (Sweke et al. 2025). Quantum convolutional networks on common image benchmarks can be simulated efficiently by a classical method based on Pauli shadows (Bermejo et al. 2026). Third, on held-out CIFAR-10, a closed-form surrogate of the PCAE2D map stays within 1 dB of matched classical feature maps. The trained circuit has not yet shown held-out generalisation. This matches large benchmarks in which simple classical models often match or beat small quantum models, and removing entanglement often costs little (Bowles et al. 2024). In PCAE the quantum processor evaluates only the circuit and returns the vector of measured Pauli expectation values. The MPS compression is quantum-inspired but runs on classical hardware, as do the Fourier positional encoding, the D_4 pooling, the decoder and the optimisation. Any quantum contribution to PCAE must therefore enter through the measured observable vector.

The restricted diagnostic of Section 4.8 is constructed so that the relevant structure is a genuine distant-element product, which raw or local features cannot represent under a fixed linear readout. That is the regime in which PCAE’s pair-observable channel separates. Natural image patches at the resolution and patch size used here

are low-entanglement, locally structured data, where a linear or shallow classical read-out is already close to sufficient. The distinction lies in the measured pair-observable channel rather than in the circuit. The ansatz uses strongly-entangling layers and does generate an entangled state, but whether the measured channel carries task-relevant information is a property of the target. On natural-image targets the pair observables add little beyond the local ones. On the constructed nonlocal target they are necessary. Together with the held-out comparisons of Section 4.3, these results delineate where the tested entangling representation does and does not yield a measurable benefit.

Three of PCAE’s ingredients recur individually in the hybrid quantum-vision literature, covering MPS compression of structured input (Stoudenmire and Schwab 2016; Han et al. 2018; Cirac et al. 2021; Guala et al. 2023), a Pauli-correlator latent space handed to a classical decoder (Havlíček et al. 2019), and a physically motivated auxiliary regulariser alongside a task loss (Shi et al. 2023). Its specific instantiation, a six-qubit VQC with a 19–27-D correlator vector and a graph-Hamiltonian term, sits within the parameter ranges typical of that literature. The measured behaviour is therefore evidence about the design pattern itself. We do not claim the boundary transfers unchanged to every dataset, patch size, or qubit count.

5.3 Topology alignment as an inductive bias

Aligning the latent interaction graph with image-patch adjacency (PCAE2D vs. PCAE1D) is an architectural inductive bias, and we test it directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives a PCAE1D-minus-PCAE2D mean PSNR difference of 0.16 dB. After averaging images within seed, the bootstrap 95% CI is $[-0.38, 0.46]$ dB and the seed-level Wilcoxon test gives $p = 0.50$. A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences, at most 0.07 dB in mean PSNR. PCAE2D completes the matched 90-step runs in roughly half the wall-clock time. At the tested scale, topology is therefore primarily a resource-cost choice rather than a reliable accuracy advantage. These tests do not establish formal equivalence.

Table 10 contrasts PCAE’s structural properties against reference frameworks, including the three nearest hybrid-vision architecture families. Its distinguishing features are a Pauli-correlator latent space, a 2D Fourier positional bias with enforced D_4 pooling, and a Hamiltonian energy term, each documented and scoped in the preceding sections.

5.4 Validated scope

- Every circuit-based reconstruction result in this paper is per-image trained. The held-out comparison of Section 4.3 uses a closed-form surrogate of the PCAE2D map. On held-out images the trained circuit stays near the random-latent range at the tested budget, so its held-out generalisation is not established.
- The per-image results of Section 4.1 are not compared with a matched-size classical per-image network, as is usual for implicit neural representations.

Table 10 Architectural differentiation from representative baseline designs, including the three nearest hybrid-vision architecture families (register-based QAE, quantum-kernel vision, TN-circuit classifier).

Axis	Conv. AE baseline	GAN baseline	Register-based QAE	Quantum-kernel vision	TN-circuit classifier	PCAE (ours)
Latent representation	Continuous $\mathbb{R}^d$	Continuous $\mathbb{R}^d$	Quantum register	Scalar kernel value	MPS/circuit state	Pauli-correlator vector
Spatial bias	Convolutional	Convolutional	Encoding-dependent	Encoding-dependent	MPS chain ordering	2D Fourier encoding
Auxiliary objective	Bottleneck / reconstruction	Adversarial loss	Fidelity / reconstruction	None (kernel value only)	Classification loss	Hamiltonian energy term
Enforced group symmetry	Not in baseline	Not in baseline	Not in reference design	Not in reference design	Not in reference design	D_4 observable pooling
Evaluated role	Reconstruction	Reconstruction	Compression	Classification	Classification	Reconstruction + diagnostics

- The D_4 -pooled map (Section 4.7) is a feature-grid-level construction and numerical diagnostic, measured on the closed-form surrogate in Table 7. Integrating it into the trainable end-to-end pipeline is future work (Section 6).
- The pair-observable necessity result (Section 4.8) is scoped to a synthetic task with closed-form features, constructed to require nonlocal structure. Ordinary natural-image reconstruction lies outside that scope (see Section 5.2).
- The hardware pilot (Section 4.4) is a five-image, unmitigated pilot with one run per image on the main backend rather than a statistically resolved hardware benchmark.
- The training-dynamics traces (Section 4.9) are an interpretability readout. The threshold fires whether the Hamiltonian term is present or absent, at 4/4 in both cases.
- The matched real-circuit topology comparison uses 3 seeds and 3 held-out images per topology, and the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation rather than formal equivalence of the topologies.
- Every result here is at a six-qubit, depth-18 scale. Trainability of variational models is known to degrade with width and depth through barren-plateau effects (McClean et al. 2018; Larocca et al. 2025), so whether the tested boundary moves at larger scale is an open question.

6 Conclusion

We presented the Pauli-Correlator Autoencoder (PCAE), a hybrid quantum-classical architecture for image reconstruction that assembles matrix-product-state patch features, a measured Pauli-correlator latent space, enforced D_4 -equivariant observable pooling, and a learnable Hamiltonian energy term into a single pipeline (Section 5.2). Across six real image datasets, per-image-trained PCAE2D models reach 25.8–41.6 dB PSNR. On held-out CIFAR-10, a closed-form surrogate of the PCAE2D observable map is equivalent within 1 dB to four resource-matched feature maps. Held-out generalisation of the trained circuit remains to be shown (Section 4.3). A hardware pilot replayed trained checkpoints on IBM Quantum devices without retraining, establishing feasibility at a five-image scope with one run per image on the main backend (Section 4.4).

Pair observables are necessary on a synthetic task requiring nonlocal structure, under a fixed linear readout (Section 4.8). The Hamiltonian energy term acts as an auxiliary training objective, and its learned energy is an interpretable readout of the trained representation (Section 5.1). The pipeline executes natively on real quantum hardware. It is also exactly D_4 -equivariant by construction, which we report as a design-correctness property rather than a quantum capability, since the same Reynolds averaging applied to a classical feature map is equally exact. At the six-qubit scale tested here, PCAE fits varied real images and runs on real quantum hardware.

Future work should integrate the D_4 -pooled map into an end-to-end trainable pipeline (Section 5.4), scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials, test held-out generalisation of the trained circuit against the same resource-matched controls, search for further real-world tasks with nonlocal structure where PCAE’s pair-observable separation translates into a reconstruction advantage, and extend the resource-matched, leakage-audited evaluation protocol to other hybrid quantum vision architectures.

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Statements and Declarations

Competing interests. The authors declare no competing interests.

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Data availability. Dataset provenance and access routes are cited in the manuscript. All datasets used (CIFAR-10, MNIST, Fashion-MNIST, and the PathMNIST, Blood-MNIST and PneumoniaMNIST subsets of MedMNIST v2) are publicly available.

Code availability. The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps each numerical headline

claim to its retained source artifact, covering the resource-matched held-out surrogate comparison and the withdrawn training-dynamics material, namely per-variant training curves of the restricted diagnostic that were interpolated from final metrics rather than logged. Automated tests verify those mappings. IBM Quantum job identifiers and reconstructed outputs are retained; the raw account-usage service response was not serialised, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

Author contributions. **Sparsho Chakraborty:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Siddhartha Patra:** Conceptualization, Supervision, Writing – review & editing, Resources. Both authors reviewed and approved the final manuscript.

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